



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD 6.GR.4

Surface Area of Pyramids

Lesson 10-5 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can find the surface area of a pyramid by adding the base area and the lateral faces.

Language Objective: I can explain my work using the words pyramid, slant height, lateral face, and base.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Pyramid

Slant height

Lateral face

Base

Apex

Lateral area



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1 A solid with a polygon base and triangular faces meeting at one point is a

2 The height of a triangular face of a pyramid, measured along the slanted side, is the

3 A triangular side face of a pyramid is a

4 The polygon on the bottom of a pyramid is the

5 The single top point where the triangular faces of a pyramid meet is the

- 6 The combined area of all the side faces, not counting the base, is the

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

For Display A (square base 6×6 , slant height 8), how did you combine the base area and the four triangular faces to get the total surface area?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Pyramid** — A solid with a flat bottom and triangle sides that meet at one point on top.
Pirámide — Un sólido con un fondo plano y lados triangulares que se unen en un punto arriba.

☐ **Slant height** — The height of a side triangle, measured along its slanted face.
Altura inclinada (apotema) — La altura de un triángulo lateral, medida por su cara inclinada.

☐ **Lateral face** — A triangle side of a pyramid, not the bottom.
Cara lateral — Un lado triangular de una pirámide, no el fondo.

☐ **Base** — The flat bottom of a pyramid.
Base — El fondo plano de una pirámide.

☐ **Apex** — The point at the top of a pyramid where the sides meet.
Ápice — El punto en la parte de arriba de una pirámide donde se unen los lados.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The base area is $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$, **and one triangular face is** $\frac{1}{2} \times 6 \times 8 = \underline{\hspace{1cm}}$.

El área de la base es $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$, y una cara triangular es $\frac{1}{2} \times 6 \times 8 = \underline{\hspace{1cm}}$.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Surface area of a pyramid is the base area plus the areas of all the triangular lateral faces.

Evidence: Students explain a square pyramid's surface area = base area + the four triangular lateral faces, and that you add every face together.

Reasoning: This evidence connects the definition of pyramid to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The base area is** $___ \times ___ = ___$, **and one triangular face is** $\frac{1}{2} \times 6 \times 8 = ___$.

El área de la base es $___ \times ___ = ___$, y una cara triangular es $\frac{1}{2} \times 6 \times 8 = ___$.

- **The total surface area is the base plus** $4 \times ___ = ___ \text{ in}^2$.

El área total de superficie es la base más $4 \times ___ = ___ \text{ in}^2$.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is** $___$.

Mi afirmación es $___$.

- **I know because** $___$.

Lo sé porque $___$.

- **So** $___$.

Entonces $___$.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Pyramid
2. **Notes 2:** Slant height
3. **Notes 3:** Lateral face
4. **Notes 4:** Base
5. **Notes 5:** Apex
6. **Notes 6:** Lateral area
7. **Watch for this mistake:** *A common mistake in Surface Area of Pyramids is finding the four triangular lateral faces correctly ($\frac{1}{2} \times \text{base edge} \times \text{slant height}$, times 4) and then stopping — forgetting to add the base area at the end. This leaves the total surface area too small by exactly the base's area. Before you submit, ask: 'Did I add the base area to the four lateral faces, or did I stop after the sides?'*
8. **Try It 1:** A. 20 in^2 — *Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 5 = 20 \text{ in}^2$. You use the slant height (5) as the triangle's height, not the pyramid's vertical height.*
9. **Try It 2:** A. 81 in^2 — *The base is a square, so base area = side $\times$ side = $9 \times 9 = 81 \text{ in}^2$. You would still add the four triangular faces to get the full surface area.*